

Ultimate DevOps Interview Guide

1. Linux User and Group Management

- Command to add a user: `sudo adduser username`
- Set user permissions: `chmod`, `chown`, `usermod`
- Real-world use: Grant sudo access, restrict access to sensitive files.

2. Docker

- Build image: `docker build -t myapp .`
- Run container: `docker run -d -p 80:80 myapp`
- Volume mounting: `docker run -v /host:/container myapp`
- Advantage: Portable environments; Disadvantage: Needs orchestration for scale

3. Bash Scripting

- Conditional: `if [ "$VAR" == "value" ]; then ...`
- Loop: `for i in {1..5}; do echo $i; done`
- Real use: Automate deployments, backups, log rotations

4. Terraform

- Define Infrastructure as Code
- Commands: `terraform init`, `plan`, `apply`, `destroy`
- Example: Provision EC2 instance with security group

5. Ansible

- Define tasks in YAML playbooks
- ad-hoc command: `ansible all -m ping`

- Playbook example: Install Nginx on target hosts

6. AWS CLI

- Create EC2: `aws ec2 run-instances ...`
- S3 upload: `aws s3 cp file.txt s3://mybucket/`
- IAM user creation: `aws iam create-user --user-name MyUser`

Interview Questions (Basic to Advanced):

Q: How do you create a user and assign to a group in Linux?

A: `useradd -G groupname username`

Q: How do you pass environment variables to a Docker container?

A: `docker run -e VAR=value image_name`

Q: How do you define a variable in Terraform?

A: `variable "region" { default = "us-east-1" }`

... [Extend with full question list]